SCHOOL OF ARCHITECTURE, COMPUTING & ENGINEERING

Submission instructions

All pages to be numbered sequentially

Module code	**CN7026**
Module title	**Cloud Computing**
Module leader	**Dr Umaima Haider**
Assignment tutor	**Dr Umaima Haider**
Assignment title	**Individual Task**
Assignment number	**1**
Weighting	**100%**
Handout date	**12th February 2026**
Submission date	**First Sit:** **Microsoft Online Course:** Week 6 (6th Mar'26) **AWS Online Labs:** Week 9 (27th March'26) **Cloud Architecture Presentation:** Week 9 (27th March'26) and Week 10 (17th April'26) **Cloud Architecture Report:** Week 12 (1st May'26)
Learning outcomes assessed by this assignment	1. Explain the core concepts of the cloud computing paradigm: how and why this paradigm shift came about, the characteristics, advantages and challenges brought about by the various models and services in cloud computing. 2. Apply the fundamental concepts in data centres to understand the trade-offs in power, efficiency and cost. 3. Discuss system virtualisation and outline its role in enabling the cloud computing system model (COI). 4. Critically assess the principles and characteristics of data centre standards (IC, COI). 5. Apply virtualization techniques and cloud computing models (DP). 6. Critically develop and evaluate solutions for real-life computational challenges (EID) 7. Assimilate new knowledge through research and be able to describe this in a written report.

	8. Group work using software packages, documentation/report writing skills and working to meet deadlines;		
Turnitin submission requirement	**YES**	**Turnitin Grade Mark feedback used?**	**No**
Moodle Grade Book submission used?	**YES**	**Moodle Grade Book feedback used?**	**Yes**
Other electronic system used?	**No**	**Are submissions / feedback totally electronic?**	**YES**
Additional information			

Form of assessment:

Individual

Number of assignment copies required:

1

Assignment to be presented in the following format:

On-line submission

Your Coursework has three main individual elements, please click on the links below to access them.

Assessment Parts	Deadline
Part 1: Microsoft Introduction to Data Centres Online Course (20%) *(You need to upload the certificates to Moodle: https://moodle.uel.ac.uk/mod/assign/view.php?id=2855696)*	**6th March 2026, 1300**
Part 2: Completion of AWS Labs demonstrating skills in cloud architecture (30%) *(No upload needed, the teaching team will be able to check your task completion via the platform)*	**27th March 2026, 1300**
Part 3: Individual Research Project Report and Presentation, 1000 - 1500 words (50%) *(You need to upload the report to Moodle: https://moodle.uel.ac.uk/mod/assign/view.php?id=2838096 , the teaching team will conduct the presentation during lab sessions).*	**Presentation: 27th March 2026; 17th April 2026** **Report: 1st May 2026, 1300**

Individual Research Project – Guide

1.0 Background

As a cloud professional, you have been allocated a scenario (see below for list) to develop a cloud-based solution for a company or an organisation.

You Can See Your Allocated Scenario here:
cn7026-202526-semb-engagement-tracker.xlsx

2.0 Course Work Requirement

For this course work, you are required to

- Start with an introduction that gives an overview of the current system in your scenario and identify a problem or issue that cloud computing can be used to address.
- You must gather functional and non-functional information about the existing system.
- Document and submit a cloud architecture for the scenario allocated to you.
- Your report must include an implementation of your architecture that has been implemented on any one of the Cloud platforms (AWS, MS Azure or Google).
- Use the information gathered to carry out the tasks and produce the deliverables in section 3.0 below
- Appendix A has some useful resources

3.0 Deliverables

You will have to produce and submit the Project report through Moodle in **PDF** format.

4.0 Documentation guidelines and other requirements

1. The *Project report* (.pdf document **max 1500** words +/10%, excluding references and pictures), which will consist of the following main areas:

The Project Report must include the following

- Cover Page
- **Title**
- **Introduction** (Write your introduction that sets the scene for the problem domain)
- **Project plan** (the plan must explain the milestones and other project planning aspects.)
- **Requirements Gathering:** Identify and specify functional and non-functional requirements to solve the presented problem
- **Data centre**
  - Type
  - Site Selection
  - Topology (Centralized/ Zoned/ Top-of-Rack/ Multi-tier)
- **Cloud Architecture:** Cloud solution
  - Choice of type of cloud platform as solution for the problem domain and rationale for choice
  - Cloud System Architecture developed to solve the problem
- **Cloud Implementation**
  - Architect the cloud solution for the scenario with a justification of the **architecture** and provide screenshots using the appropriate cloud platform
- **Costing**
- **Analysis and Reflection**
  - Present your analytical reflection on how to maintain the system, manage its evolution, respond to the evolving business needs as well as regulations and government rule
- **References**

2. Presentation Template: COURSEWORKPRESENTATIONTEMPLATE.

5.0 Marking Scheme

Marking component	Marks	What is needed?
Implementation	30%	● Cloud architecture best practices followed. ● Screenshots demonstrating architecture provided
Design	50%	● Project Plan ● Problem analysis & Requirement specification ● Rationale for choice of data centre, services and scope of application. ● Notes about the progress of the application. ● What has been finished; what is left to do; what needs further work. ● Evaluation/critical appraisal A reflective piece about what the prototype has shown and whether it meets its needs.
Oral Presentation	10%	● Present the system summary and cloud architecture diagram ● Clear and concise verbal communication
Documentation Presentation	10%	● Coherent structure, including separate sections and a table of contents. ● Good use of English (check for spelling mistakes) ● All images in the report appropriately scaled ● Good use of appropriate references (including labelling).

6.0 Marks Breakdown

Implementation (30 Marks)

	None (0 Marks)	Some features (0-15 Marks)	50% (30 Marks)
50% of application implemented and showcased in report			

Design (50 Marks)

	Not present (0 Marks)	Briefly explained (3 Marks)	Adequately explained (6 Marks)	In depth with evidence of understanding on how to apply technique (10 Marks)
Project Plan				
Problem analysis & Requirement specification				
Rationale for choice of services and scope of application.				
Choice of Data centres explained				
Costing of application				
Evaluation/critical appraisal A reflective piece about what the prototype has				

shown and whether it meets its needs.				

ORAL PRESENTATION (10 Marks)

	Poor (0 Marks)	**Average (1 Marks)**	**Good (5 Marks)**
Well-designed, relevant introduction and logically correct architecture diagram			
Present confidently with clear speech, effective body language, and good time management.			

DOCUMENTATION (10 Marks)

	Poor (0 Marks)	**Average (1 Marks)**	**Good (2.5 Marks)**
Coherent structure, including separate sections and a table of contents			
Good use of English (check for spelling mistakes)			
All images in the report appropriately scaled			
Good use of appropriate references (including labelling).			

7.0 Further notes on the coursework

The following guidelines will help you to achieve a good mark for your assignment:

- Your report must include glossaries, an abstract or an executive summary.
- You should focus on strictly covering the marking components for this assignment.
- Appendices and references will not be included in your word count.

What to submit

Your submission should be organized in these files and folders:

- Documentation
- Make sure that you submit the correct files, that they are complete and not corrupt. Test your submission by downloading your files to a new directory on your computer and make sure that everything will work as expected. Check that you can open all your files, and that the files are not corrupted. Under no circumstances will markers attempt to restore files from defective/corrupted files. If any part of your assignment cannot be assessed because of file corruption you will score zero marks for that component.
- Please do not submit files that are not part of your assignment (e.g., extra files in your prototype folder or different copies of your assignment).

English Language Support

You can get English Language support from our English Academic Support Tutor; they can only help you with grammar and writing and not the technical content.
https://uelac.sharepoint.com/sites/ACE-GettingStarted/SitePages/Academic-Support.aspx

Referencing

https://www.uel.ac.uk/study/student-life/library-archives-learning-services/info-skills-guides/referencing-information

Plagiarism

A simple definition:
Plagiarism: a piece of writing that has been copied from someone else and is presented as being your own work (WordNet ® 1.6, © 1997 Princeton University).
Plagiarism will not be tolerated in any way and will be dealt according to UEL policies.

Appendix A: Useful Resources

1- AWS Whitepapers and Guides https://aws.amazon.com/whitepapers/

Here you can find a lot of various examples such as Airline Information Management System Implementation on AWS. Such diagrams are beneficial in different ways as they tell how to implement scalable systems using AWS.

2- The 5 Pillars of the AWS Well-Architected Framework which provides you will good design principles and best practices. It is also supported with a well-designed FAQs. The page can be accessed through the following link: The 6 Pillars of the AWS Well-Architected Framework | AWS Partner Network (APN) Blog (amazon.com)

3 - Innovative cloud architectures from AWS partners and customers: https://aws.amazon.com/architecture/this-is-my-architecture/

4 - How to use the AWS Simply Month Calculator

- Getting Started with AWS Simple Monthly Calculator - Estimate Cloud Pricing on AWS - YouTube
- Simple Monthly Calculator: https://calculator.s3.amazonaws.com/index.html

5 - To implement the architecture in a diagrammatic way for this you can use these links

- Online AWS Architecture Diagram Tool (visual-paradigm.com)
- AWS Architecture Diagram Tool | Draw AWS Diagrams | Creately
- AWS Architecture Icons (amazon.com)